Miguel Salvacion

msalvacion@cmu.edu | miguelsalv.com | linkedin.com/in/msalvacion | github.com/miguel-salv

Education

Carnegie Mellon University Pittsburgh, PA
B.S. and M.S. in Electrical and Computer Engineering — GPA: 3.81/4.00 — Dean's List *May 2027*

Leadership: Filipino Student Association Treasurer – Managed $9,000 budget for 150+ member organization

Technical Skills

Languages: C/C++, ARM Assembly (ARMv7-M Thumb-2), Python, SystemVerilog, MATLAB
Protocols & Interfaces: CAN, I2C, SPI, UART, PWM, ADC, DMA
Embedded & RTOS: Zephyr RTOS, STM32 (Cortex-M4), Teensy 4.1, Linux, GDB, OpenOCD, JTAG
Hardware & EDA: KiCad, Cadence Virtuoso, Vivado, FPGA, SolidWorks, Autodesk Fusion, PCB Design

Work Experience

Hardware Systems Engineering Intern | *eDrive Units, Diagnostics* Summer 2025 & 2026
Ford Motor Company *Detroit, MI*

- Resolved a 5.8 V drop across an EV control unit via schematic review, identifying elevated temperatures and undersized wiring as root causes and specifying upgraded wire gauges and routing changes.
- Analyzed field warranty telemetry and hardware teardowns across EV platforms (Mach-E, E-Transit), isolating a mechanical park-system ratcheting fault and presenting an 8D root-cause analysis for technical review.
- Authored secondary electric drive unit teardown and service procedures, validating modular eMotor sub-assembly repair to eliminate full-unit replacements in the field.

Embedded Systems Researcher | *Semiconductor Fab, RF Hardware* Jan 2026 – Present
CMU Hacker Fab *Pittsburgh, PA*

- Designed and built an open-source, automated RF impedance matching network for $636 (80%+ cost reduction over commercial units), eliminating manual tuning during thin-film depositions.
- Achieved a 1.2 VSWR at 95–100 W and sustained a stable plasma for 60+ minutes by implementing real-time coordinate descent on a Teensy 4.1 driving air-variable capacitors.
- Designed a custom control PCB in KiCad and verified SWR sensing and power-protection circuits in ngspice, confirming ADC inputs stayed below 3.3 V across tolerance and parasitic corner simulations.

Head Teaching Assistant | *RTOS, Firmware, ARM Architecture* Jan 2026 – Present
CMU 18-349: Introduction to Embedded Systems *Pittsburgh, PA*

- Managed 10 TAs and mentored 60+ students debugging custom preemptive RTOS kernels, resolving PendSV context-switch faults, RMS bugs, and HLP priority-inversion edge cases using GDB and OpenOCD over JTAG.
- Guided firmware driver development on STM32 Cortex-M4, including MMIO register configuration, interrupt-driven serial protocols, and hardware PWM to integrate sensors and actuators.
- Conducted design reviews for student 2-layer PCBs and supported hardware-software integration across closed-loop PID motor tuning, RMS thread scheduling, and sensor-actuator interfacing.

Projects

Distributed Drive-by-Wire System | *CAN Bus, Zephyr RTOS, RAFT, PCB Design* Aug 2026 – Present

- Implemented a 3-node distributed vehicular network on Zephyr RTOS, meeting <50 ms end-to-end latency requirements for real-time steering and throttle actuation.
- Architected dual-redundant CAN channels with frame deduplication, channel voting, and RAFT consensus.
- Designed zone controller PCBs in KiCad and synchronized distributed virtual clocks to coordinate closed-loop PID motor velocity control.

Preemptive RTOS & Vehicle Platform | *ARM Assembly, C, RMS, Fusion 360* Aug 2025 – Dec 2025

- Built a preemptive real-time kernel for STM32 in C and ARM Assembly, implementing PendSV context switching, newlib SVC syscalls, and Highest Locker Protocol mutexes.
- Implemented Rate-Monotonic Scheduling to ensure zero deadline misses across concurrent closed-loop PID speed control, UART telemetry, and I2C LCD tasks.
- Designed a 2-layer PCB in Fusion and wrote bare-metal drivers for quadrature decoding, PWM, UART, and I2C.